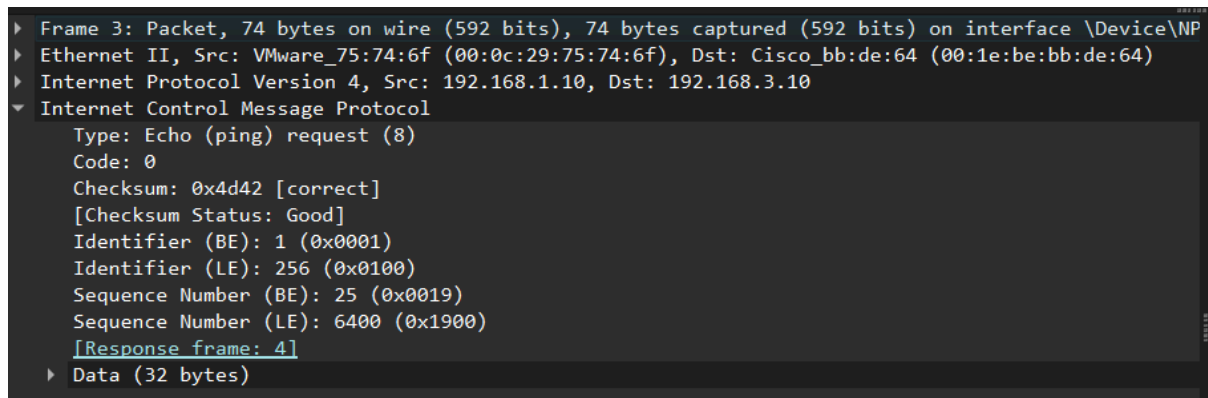


Lab Report - Build and analyse a Network with Wireshark

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In this lab we conducted research regarding the implementation of a real world Network. For this we used actual hardware components. The main objective of this lab was to deepen our understanding of network protocols, as well as to operate analysis tools. After the initial set up of our network topology, we had to use wireshark to analyse the network protocols. The result of this lab was a wireshark capture sheet of a ping. In the following we will explain how the process of pinging and traceout works at the protocol level. We also will briefly highlight the main differences between RIPv1 and RIPv2. Our appendix includes a set of questions, which were answered using our wireshark file and the corresponding RFCs (Request for Comments).

The ping operates entirely within the network layer. When the ping is sent by typing the ip address of the destination device into the terminal, it first starts with an ICMP Echo (ping) request. After this an 74 Echo ping reply is sent by the other device. This is followed by an interchange of request and replies of the two ip addresses. When the source receives the Echo Reply, it uses it to calculate the Round-Trip Time.

A screenshot of the Wireshark packet capture details pane. The selected packet is Frame 3, an ICMP Echo (ping) request. The details show the Ethernet II header with source VMware_75:74:6f and destination Cisco_bb:de:64. The Internet Protocol Version 4 header shows source 192.168.1.10 and destination 192.168.3.10. The Internet Control Message Protocol section shows it is an Echo (ping) request with code 0, a correct checksum of 0x4d42, identifier 1, and sequence number 25. A link to the response frame (4) is provided. The data field contains 32 bytes.

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▶ Frame 3: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NP
▶ Ethernet II, Src: VMware_75:74:6f (00:0c:29:75:74:6f), Dst: Cisco_bb:de:64 (00:1e:be:bb:de:64)
▶ Internet Protocol Version 4, Src: 192.168.1.10, Dst: 192.168.3.10
▼ Internet Control Message Protocol
  Type: Echo (ping) request (8)
  Code: 0
  Checksum: 0x4d42 [correct]
  [Checksum Status: Good]
  Identifier (BE): 1 (0x0001)
  Identifier (LE): 256 (0x0100)
  Sequence Number (BE): 25 (0x0019)
  Sequence Number (LE): 6400 (0x1900)
  [Response frame: 4]
▶ Data (32 bytes)
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The traceroute is used to determine the path packets take to reach a destination. The purpose of traceroute is to prevent packets from wandering in endless routing loops. This is achieved by using the Time-to-live field in the IP header. Every router that forwards a packet lowers this number by 1. If the number reaches 0, the router drops the packets and sends back an error message. The first packet starts with TTL at 1. If this gets dropped, it sends another packet with TTL at 2. This continues until a packet reaches the actual destination host.

The main difference between RIPv1 and RIPv2. RIPv1 is a classful routing protocol. It does not send subnet mask information in its routing updates. RIPv2 is a classless routing protocol with subnet mask. RIPv1 uses broadcast, while RIPv2 is multicast, only to a specific address. Another aspect is security. RIPv1 has no authentication mechanism. RIPv2 uses authentication, plain text and encryption.

Appendix

ICMP echo request - echo reply (ping)

- 1. In which RFC is ICMP defined?**
ICMP is defined in RFC 792.
- 2. The Destination-MAC-Address of the echo request packet is the MAC-Address of which device?**
It is the MAC address of the next-hop device.
- 3. Can the PC1 see the MAC-Address from PC3? If not, explain why.**
It cannot, this is due to MAC addresses only having local significance. When a packet crosses a router to reach another network, the original Layer Ethernet header is replaced.
- 4. How does the ethernet layer signal that it is transporting IP data?**
Through the EtherType field in the Ethernet II frame header.
- 5. How does the IP layer signal that it is transporting ICMP data?**
Through the protocol field in the IPv4.
- 6. How many bits has the type field of the ICMP header?**
The type field is 8 bits long.
- 7. Which type number has an echo request packet?**
Type 8.
- 8. Which type number has an echo reply packet?**
Type 0.
- 9. What data must be in the data field of an echo reply message? (Search for an answer in the RFC for ICMP.)**
RFC 792 tells us that the data received in the Echo Request message must be returned unaltered in the reply.

Traceroute

- 1. What transport protocol uses traceroute for its packets?**
ICMP.
- 2. What destination port uses traceroute for its first packet?**
It uses 33434.
- 3. What is the purpose of the “time to live” field in the IP header? (Search for an answer in the RFC for IP.)**
Works as a mechanism to prevent packages from looping endlessly.
- 4. What is the first “time to live” number used by traceroute?**
The first packet is sent with a TTI of 1.
- 5. Why is the router R1X sending a time-to-live exceeded message, as a response to the first packet of the traceroute program?**

3. How is a Multicast-MAC-Address generated?

The first 24 bits are fixed (01:00:5E). The 25th bit is always 0. The remaining 23 bits are mapped directly from the lower 23 bits of the multicast IP address.

4. Which IP-Address space is reserved for multicast?

The Class D network range: 224.0.0.0 to 239.255.255.255 (or 224.0.0.0/4).

5. What Networks does R2 propagate to R1 using RIP?

R2 is directly connected to the 192.168.2.0/24 transit network and the 192.168.3.0/24 local network. R2 propagates the 192.168.3.0 network.

6. What extra fields are found in RIPv2 compared to RIPv1?

RIPv2 introduces the Subnet Mask, Next Hop, and Route Tag fields, alongside supporting authentication headers within the first entry slot.

7. Why is the Address in the Next-Hop Field 0.0.0.0? (Search for an answer in the RFC for RIPv2.)

As stated in RFC 2453, a Next-Hop of 0.0.0.0. Indicates that the originator of the routing update is itself the best next hop.

8. Why is RIPv2 switched to the Multicast-address 224.0.0.9. (Search for an answer in the RFC for RIPv2.)

To reduce processing overhead on non-routing hosts. This is different to RIPv1 broadcasting to every machine on the subnet.